Build a Windows Server Active Directory environment to practice domain administration, Group Policy management, delegated administration, and NTFS/share permissions

Active Directory Home Lab Setup

Meet Patel

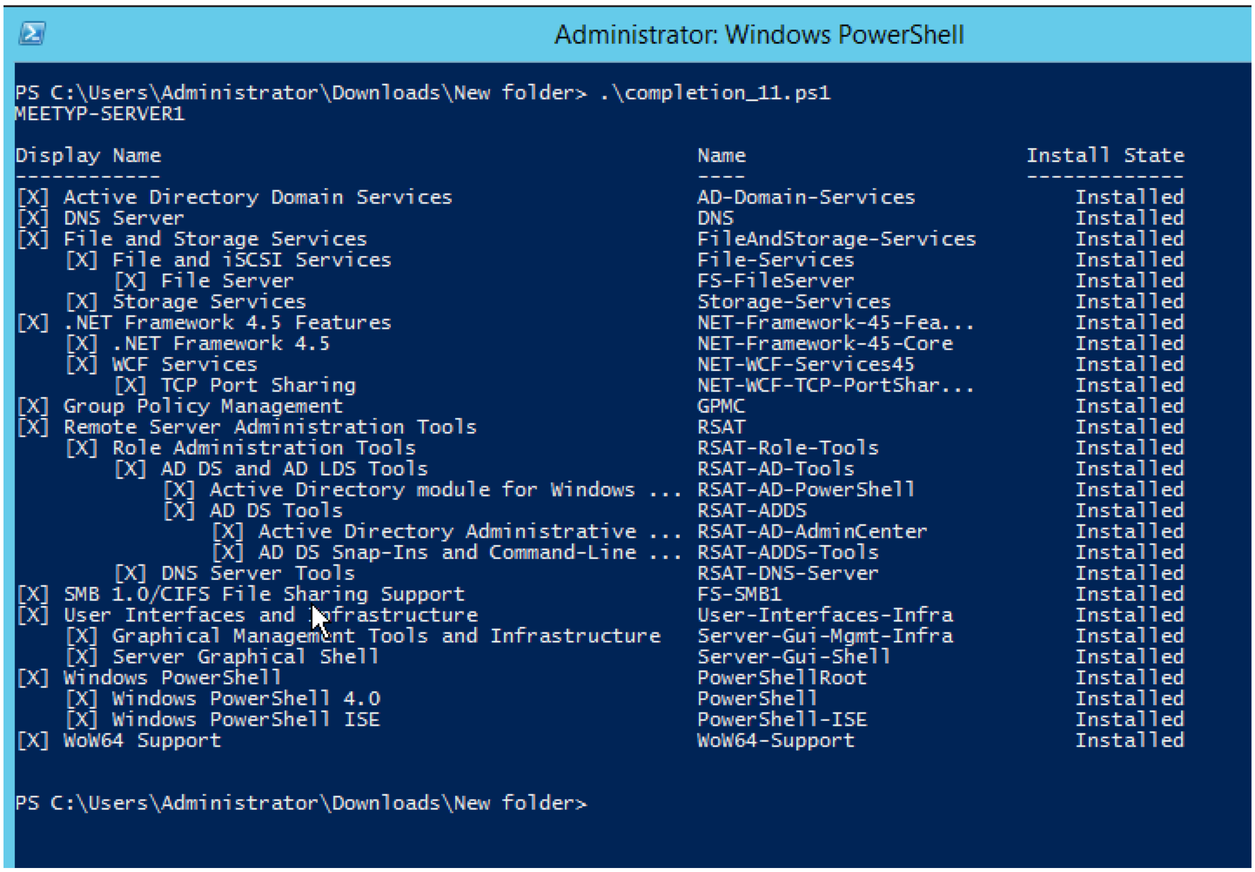
I first configured the server with the hostname MEETYP-Server1.

I then configured the network using NAT so that the VM could access the internet while also allowing communication between the host machine and the VM.

Once the network was set up, I fully patched the server by installing the latest Windows updates.

After the updates were completed, I promoted the server to a Domain Controller using Server Manager. During the promotion process, I created a new Active Directory forest and domain named MEETYP.

I used PowerShell to verify the server hostname and check which Windows Server roles and features were installed.



```
Administrator: Windows PowerShell

PS C:\Users\Administrator\Downloads\New folder> .\completion_11.ps1
MEETYP-SERVER1

Display Name                                            Name                       Install State
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[X] Active Directory Domain Services                    AD-Domain-Services             Installed
[X] DNS Server                                          DNS                            Installed
[X] File and Storage Services                           FileAndStorage-Services        Installed
    [X] File and iSCSI Services                         File-Services                  Installed
        [X] File Server                                 FS-FileServer                  Installed
    [X] Storage Services                                Storage-Services               Installed
[X] .NET Framework 4.5 Features                         NET-Framework-45-Fea...        Installed
    [X] .NET Framework 4.5                              NET-Framework-45-Core          Installed
    [X] WCF Services                                    NET-WCF-Services45             Installed
        [X] TCP Port Sharing                            NET-WCF-TCP-PortShar...        Installed
[X] Group Policy Management                             GPMC                           Installed
[X] Remote Server Administration Tools                  RSAT                           Installed
    [X] Role Administration Tools                       RSAT-Role-Tools                Installed
        [X] AD DS and AD LDS Tools                      RSAT-AD-Tools                  Installed
            [X] Active Directory module for Windows ... RSAT-AD-PowerShell             Installed
            [X] AD DS Tools                             RSAT-ADDS                      Installed
                [X] Active Directory Administrative ... RSAT-AD-AdminCenter            Installed
                [X] AD DS Snap-Ins and Command-Line ... RSAT-ADDS-Tools                Installed
        [X] DNS Server Tools                            RSAT-DNS-Server                Installed
[X] SMB 1.0/CIFS File Sharing Support                   FS-SMB1                        Installed
[X] User Interfaces and Infrastructure                  User-Interfaces-Infra          Installed
    [X] Graphical Management Tools and Infrastructure   Server-Gui-Mgmt-Infra          Installed
    [X] Server Graphical Shell                          Server-Gui-Shell               Installed
[X] Windows PowerShell                                  PowerShellRoot                 Installed
    [X] Windows PowerShell 4.0                          PowerShell                     Installed
    [X] Windows PowerShell ISE                          PowerShell-ISE                 Installed
[X] WoW64 Support                                       WoW64-Support                  Installed

PS C:\Users\Administrator\Downloads\New folder>
```

Figure 1: PowerShell verification of server hostname and installed Windows Server roles and features

Script: $env:computername
Import-module servermanager ; Get-WindowsFeature | where-object {$_.Installed -eq $True}

I used PowerShell to view the computer accounts in Active Directory and verify their computer names, IPv4 addresses, operating systems, and service pack information.

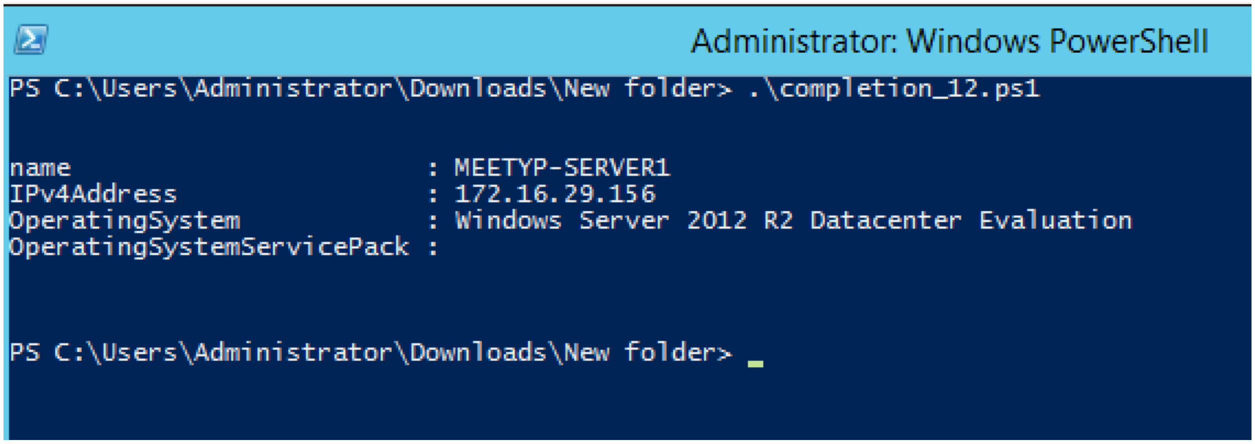


Figure 2: PowerShell verification of Active Directory computer accounts and system information

Script: Get-ADComputer -Filter * -Properties ipv4Address, OperatingSystem, OperatingSystemServicePack | Format-List name, ipv4*, oper*

Kiosk OU and User Account

I created a new Organizational Unit (OU) called Kiosk in Active Directory. I then created a user account named kiosk, which will be used to test the kiosk workstation settings.

Creating the Kiosk Group Policy

I opened the Group Policy Management Console (GPMC) and created a new Group Policy Object (GPO) called Kiosk Policy. I linked the GPO to the Kiosk OU so that the policy would apply to the users and computers placed in this OU.

I configured settings under both Computer Configuration and User Configuration to help create a more restricted and secure kiosk environment. The goal was to limit access to system settings and administrative functions while still allowing the kiosk user to perform basic tasks such as accessing the internet.

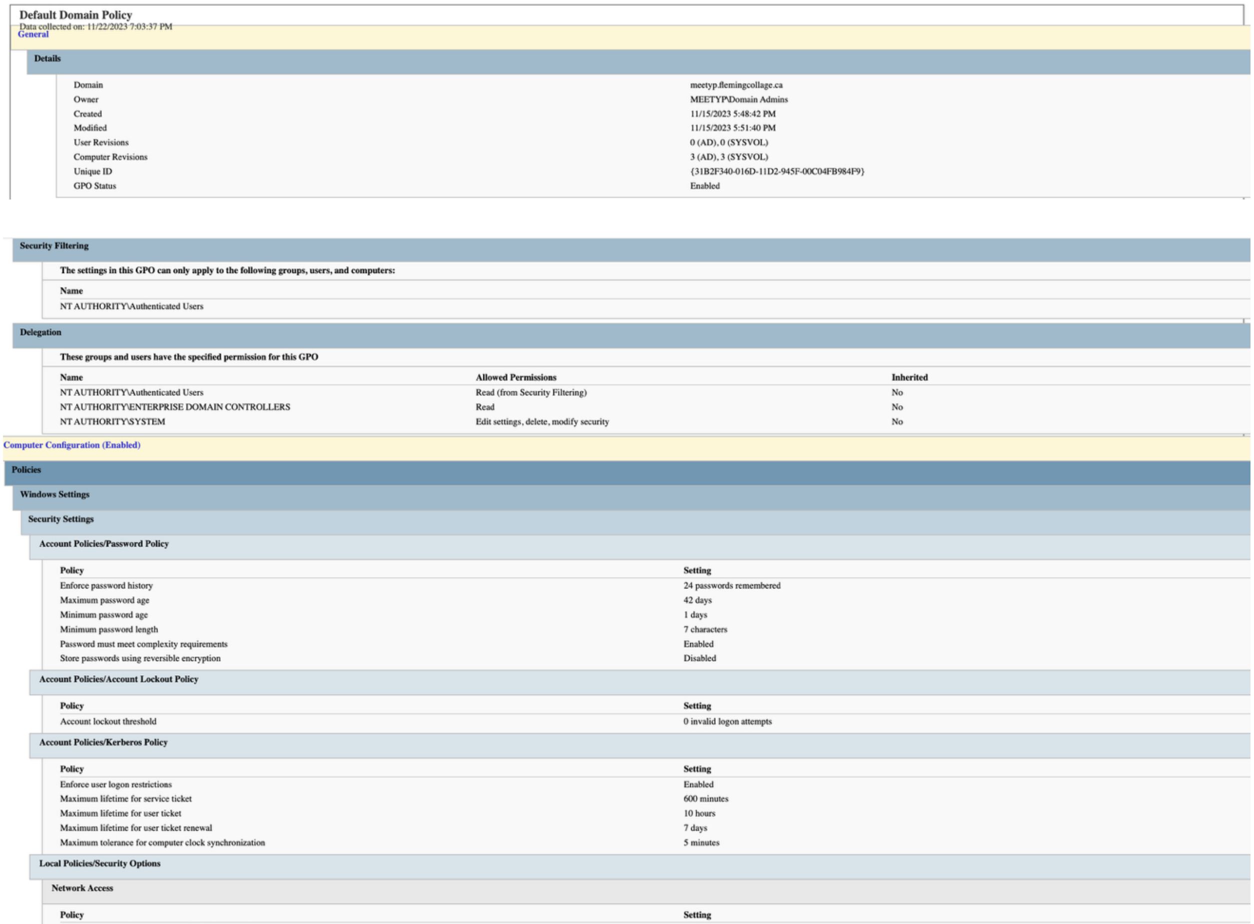
Testing the Policy

After configuring the GPO, I logged in using the kiosk user account to test the settings. I checked that the restrictions were being applied as expected and that the user could still access the necessary functions.

I made adjustments to the Group Policy settings where needed and tested the configuration again to make sure the kiosk environment was working properly.

GPO Export

Once I was satisfied with the configuration and testing, I exported the Kiosk Policy GPO. This provides a backup of the policy and allows the configuration to be reviewed or imported into another environment if needed.



Default Domain Policy
Data collected on: 11/22/2023 7:03:37 PM

General

Details

Domain	meetyp.flemingcollage.ca
Owner	MEETYP\Domain Admins
Created	11/15/2023 5:48:42 PM
Modified	11/15/2023 5:51:40 PM
User Revisions	0 (AD), 0 (SYSVOL)
Computer Revisions	3 (AD), 3 (SYSVOL)
Unique ID	{31B2F340-016D-11D2-945F-00C04FB984F9}
GPO Status	Enabled

Security Filtering

The settings in this GPO can only apply to the following groups, users, and computers:

Name
NT AUTHORITY\Authenticated Users

Delegation

These groups and users have the specified permission for this GPO

Name	Allowed Permissions	Inherited
NT AUTHORITY\Authenticated Users	Read (from Security Filtering)	No
NT AUTHORITY\ENTERPRISE DOMAIN CONTROLLERS	Read	No
NT AUTHORITY\SYSTEM	Edit settings, delete, modify security	No

Computer Configuration (Enabled)

Policies

Windows Settings

Security Settings

Account Policies/Password Policy

Policy	Setting
Enforce password history	24 passwords remembered
Maximum password age	42 days
Minimum password age	1 days
Minimum password length	7 characters
Password must meet complexity requirements	Enabled
Store passwords using reversible encryption	Disabled

Account Policies/Account Lockout Policy

Policy	Setting
Account lockout threshold	0 invalid logon attempts

Account Policies/Kerberos Policy

Policy	Setting
Enforce user logon restrictions	Enabled
Maximum lifetime for service ticket	600 minutes
Maximum lifetime for user ticket	10 hours
Maximum lifetime for user ticket renewal	7 days
Maximum tolerance for computer clock synchronization	5 minutes

Local Policies/Security Options

Network Access

Policy	Setting

Figure 3: Exported Kiosk Policy Group Policy Object (GPO) backup

Creating the Head Office OU

Using Active Directory Users and Computers, I created a new OU called Head Office.

Inside the Head Office OU, I created the following groups:

- IT
- HR
- Finance
- Head Office Users
- Managers

I then created the following user accounts:

- bobsmith
- maryhairnet
- sallymanager

- me (My own user account)

I added bobsmith to the HR group and maryhairnet to the Finance group. I added my own account to the IT group.

I also created the sallymanager account and added the user to both the HR and Managers groups.

Finally, I added all of the users created in the Head Office OU to the Head Office Users group.

Creating the Montreal OU

I then created a new top-level OU called Montreal.

Inside the Montreal OU, I created a group called Montreal Users and created the dunns user account. I added dunns to the Montreal Users group.

I also created the joedonuts user account and added the user to the Montreal Users group and a new group called Montreal-IT. In addition, I added joedonuts to the Server Operators group.

After completing these steps, I verified the users, groups, and group memberships in Active Directory Users and Computers.

Delegating Control of the Montreal OU

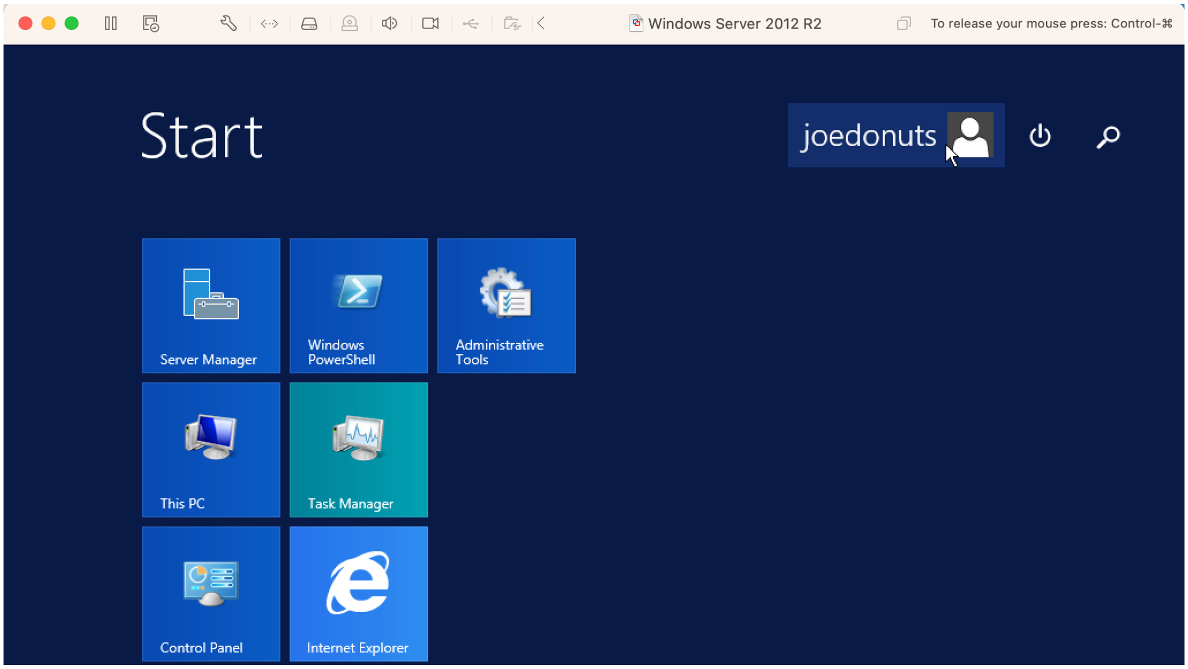
Using Active Directory Users and Computers, I delegated control of the Montreal OU to the Montreal-IT group. This allows members of the group to perform specific administrative tasks within the OU without giving them full domain administrator permissions.

Testing the Delegation

I logged into the server using the joedonuts account, which is a member of the Montreal-IT group.

After logging in, I opened Active Directory Users and Computers and attempted to reset the password for the bobsmith user account.

The password reset attempt was made to verify whether the delegated permissions for the Montreal-IT group allowed Joe to modify users outside of the Montreal OU.



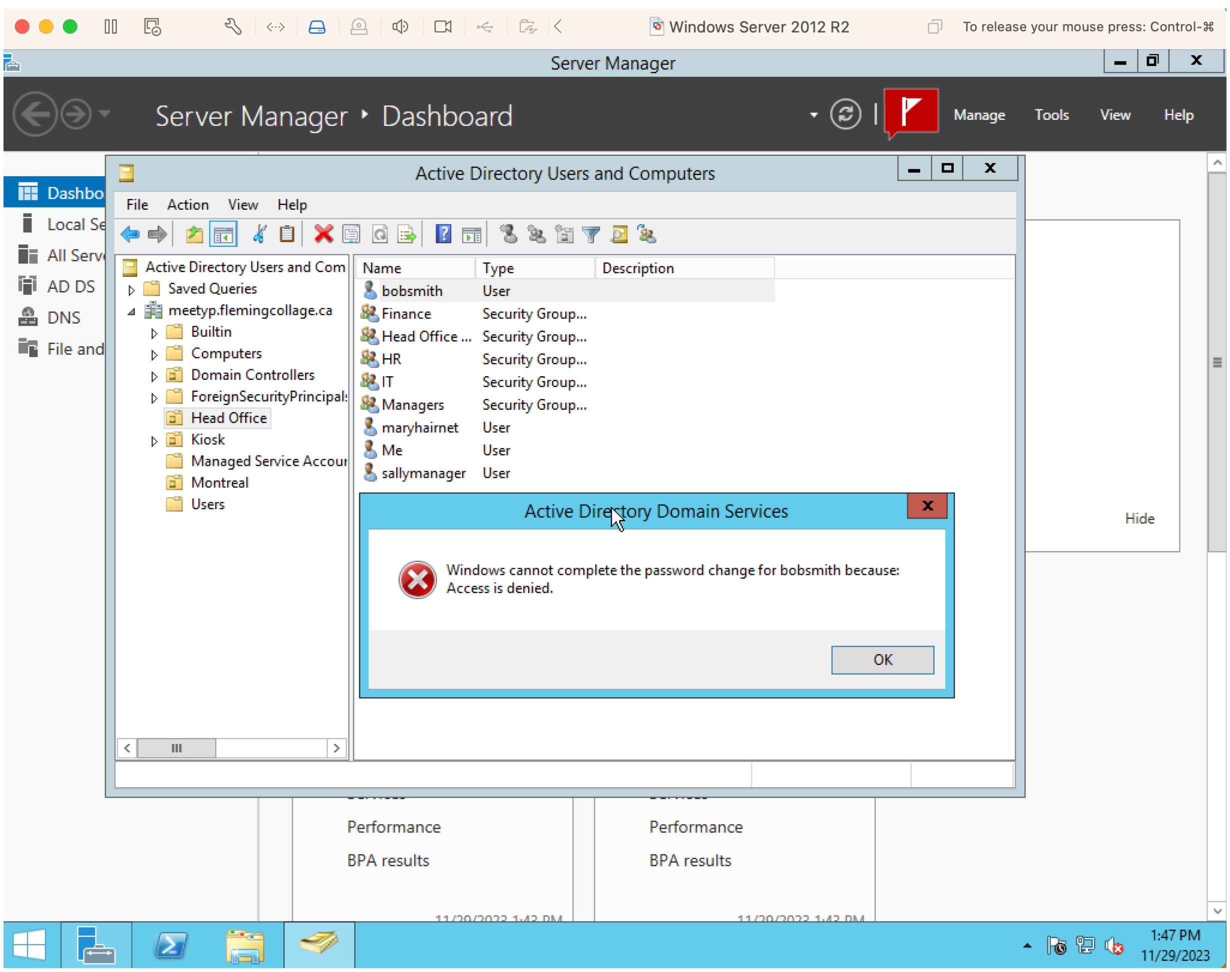


Figure 4: Delegated permissions test showing access denied when attempting to modify a user outside the delegated OU

I recorded a screenshot of the resulting message, including the Active Directory Users and Computers window, as evidence of the test result.

Creating the Managed Groups OU

I created a new top-level Organizational Unit (OU) called Managed Groups in the root of the domain.

Inside the OU, I created two security groups:

- **HR_Share_RO**
- **HR_Share_ReadWrite**

I added maryhairnet to the HR_Share_ReadWrite group and also added her to the Server Operators group.

Creating and Sharing the Folder

On the server, I created a folder named SHARE on the C:\ drive and shared it over the network. I configured the share permissions to give Everyone full control.

Inside the SHARE folder, I created an HR folder. Within the HR folder, I created another folder called HR Managers.

Configuring Folder Permissions

I configured the permissions on the HR folder as follows:

- **HR_Share_RO** – Read-only access
- **HR_Share_ReadWrite** – Full Control

I then disabled inheritance on the HR Managers folder and configured its permissions so that only the Managers group had Full Control, as required.

Testing the Permissions

After configuring the permissions, I logged into the server using the maryhairnet account and attempted to access the HR Managers folder.

Since Mary is a member of the HR_Share_ReadWrite group, but the HR Managers folder has inheritance disabled and only grants access to the Managers group, I expected the access attempt to be denied.

I recorded a screenshot of the resulting access message as evidence of the permissions test.

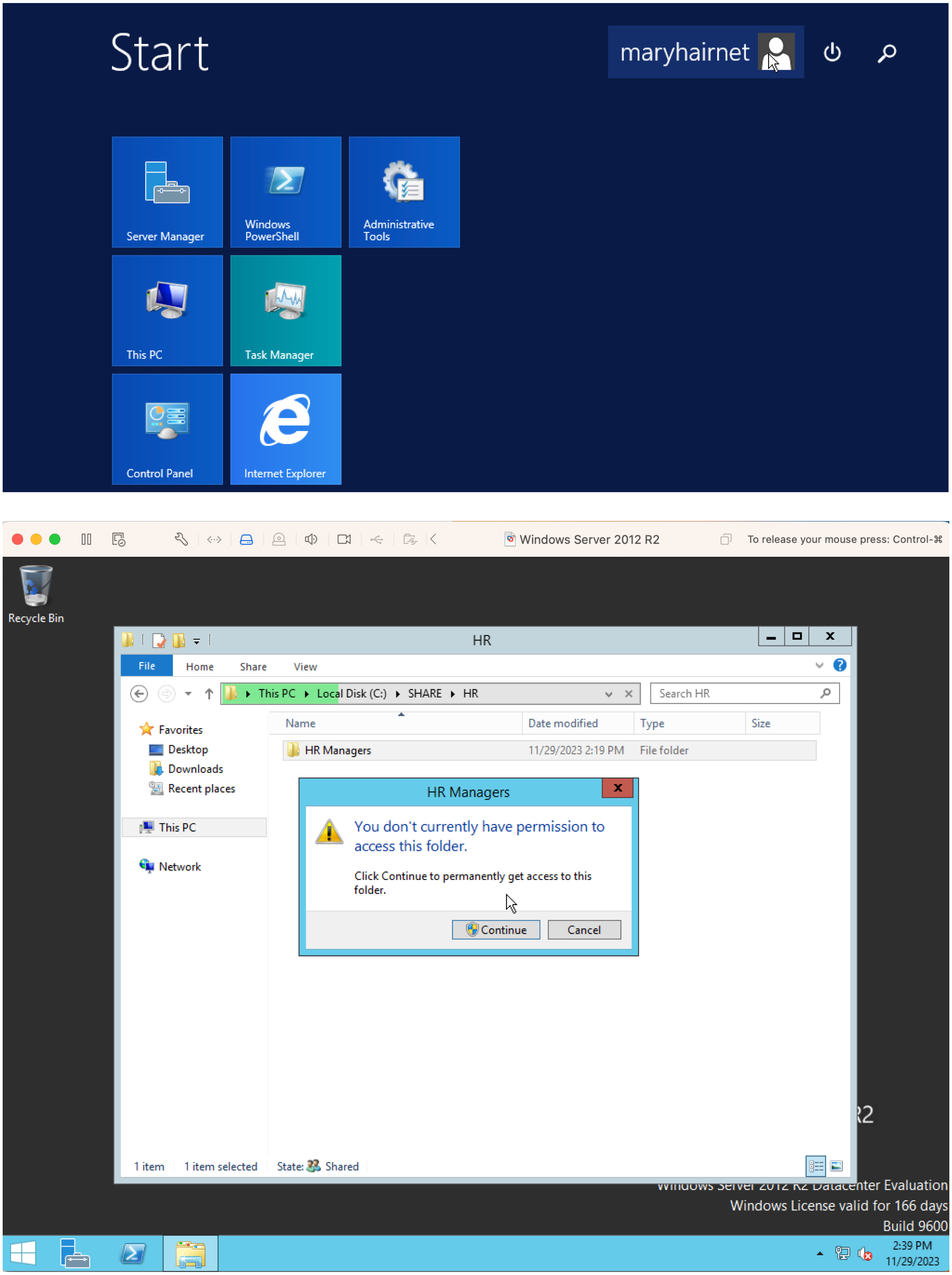
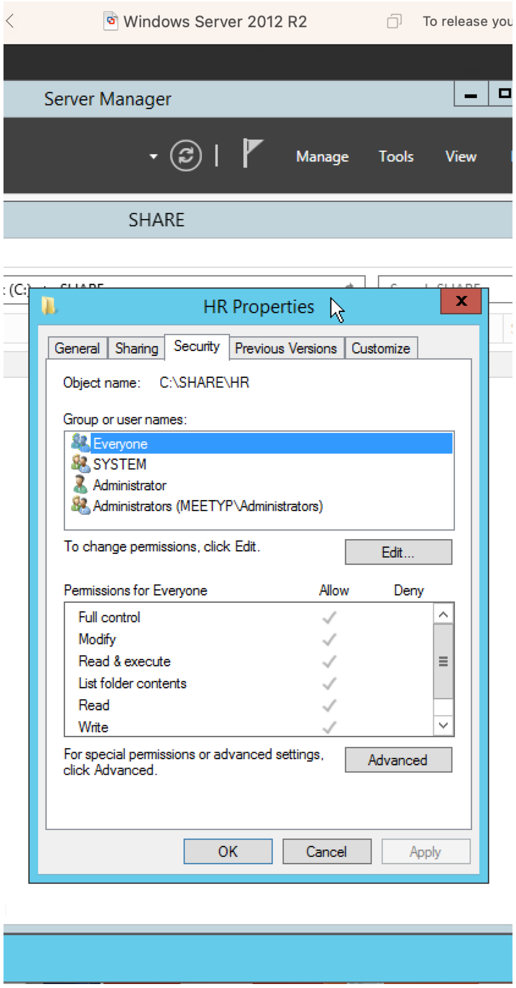


Figure 5: Access control test showing denied access to the HR Managers folder for the maryhairnet account

The following screenshots show the NTFS access control lists (ACLs) configured on the HR and HR Managers folders.



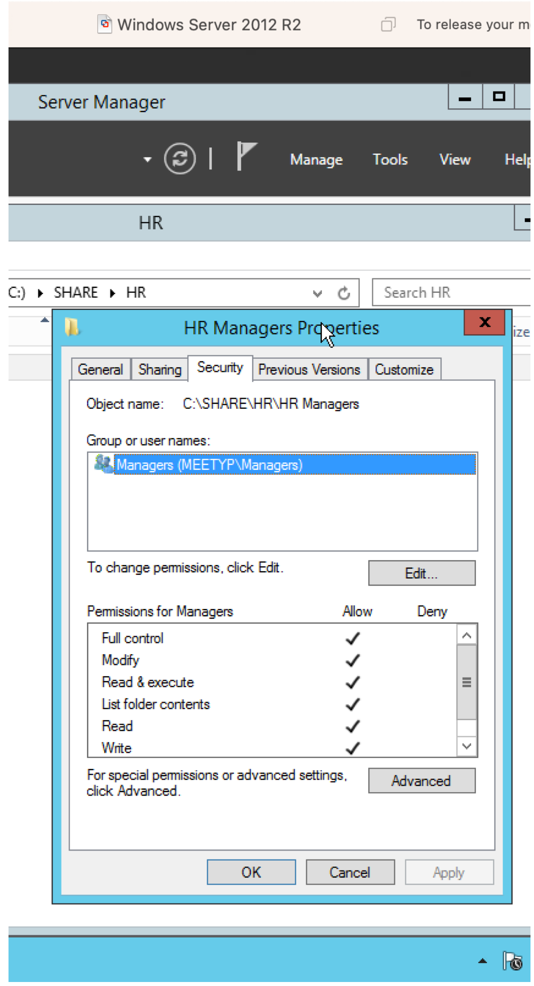


Figure 6: NTFS permissions configured on the HR folder and HR Managers folder